

Physics A (H156, H556)

Exploring - Radioactivity ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **90**.
- The total number of marks may take into account some 'either/or' question choices.

- 1(a) Technetium-99m ($^{99}_{43}\text{Tc}^m$) is a metastable radioisotope which can be used as a tracer in medical diagnosis. It is injected into the body and decays by gamma emission into technetium-99 according to the following chemical equation.



- i. A technetium-99m tracer with an activity of 900 MBq is injected into a body. The half-life of technetium-99m is 6.01 hours.

Calculate the number of technetium-99m nuclei initially present in the tracer.

number = [3]

- ii. Calculate the time in hours taken for the activity of the tracer to have fallen to 3.0% of its initial activity.

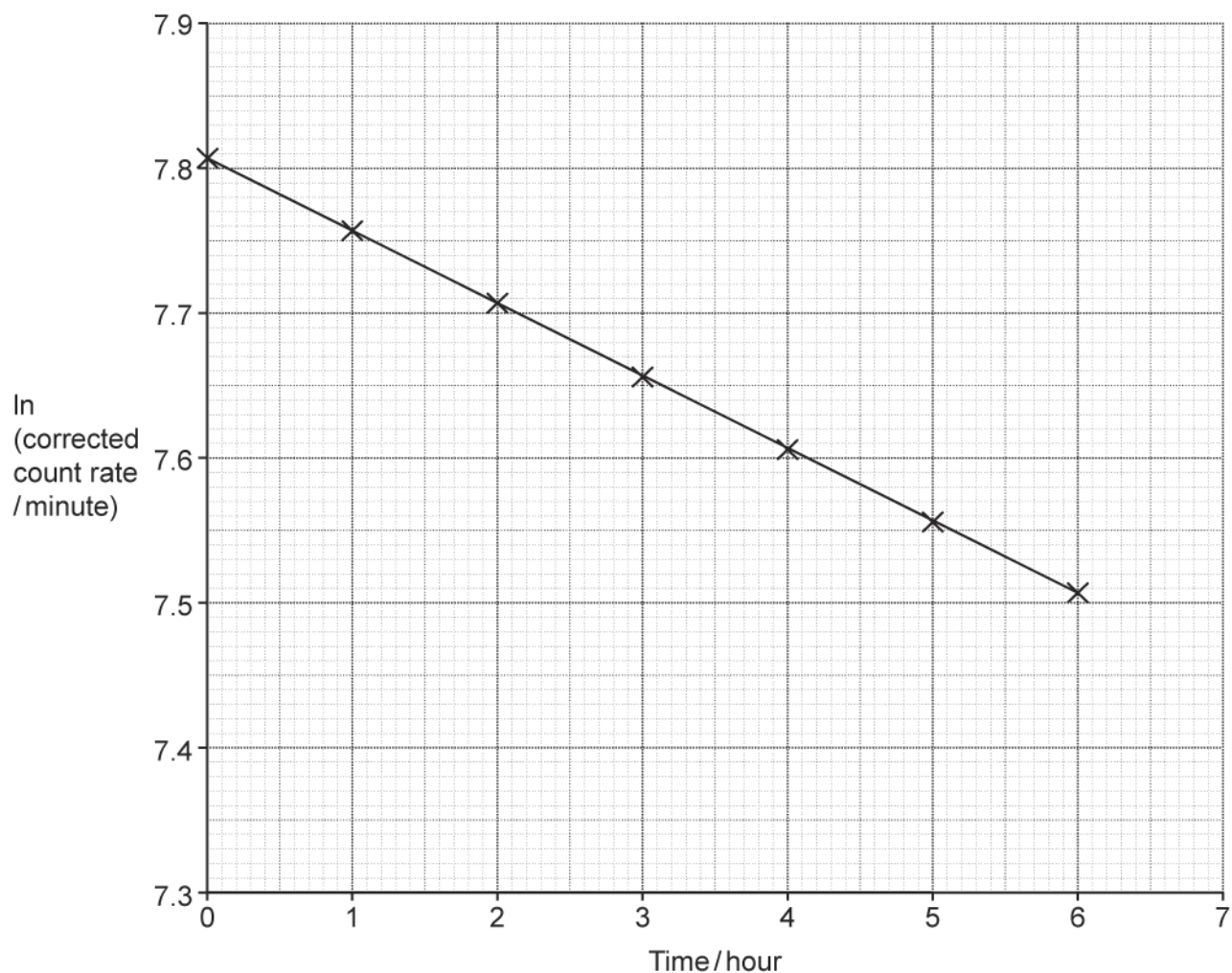
time = hours [3]

- (b) The daughter nucleus ($^{99}_{43}\text{Tc}$) decays by beta emission with a half-life of a little over 200 000 years. Approximately 50% of it is stored in the bones, and 50% is passed out of the body.

Suggest why the presence of this remaining $^{99}_{43}\text{Tc}$ in the body causes little additional risk to the patient.

 ----- [1]

- (c) The half-life of a radioisotope is to be determined using suitable apparatus. Each count represents one decay. The number of counts is measured for one minute every hour over a period of 6 hours. When the data has been collected, a graph of $\ln(\text{corrected count rate / minute})$ against time is plotted and shown below.



- Describe an appropriate method that could be used to obtain this data, naming any apparatus and safety precautions taken.
- Use the graph given above to determine the half-life of this radioisotope showing clear working.

----- [6]

2 This question is about the rate of decay of a radioactive source.

Which of the following statements is/are true?

The rate of decay is

- 1 dependent on the decay constant.
- 2 independent of the mass of the source.
- 3 dependent on time.

- A 1 only
- B 1 and 3
- C 2 only
- D 2 and 3

Your answer

☐

[1]

3(a) Radiographers commonly use molecules containing fluorine F-18 as tracers in positron emission tomography (PET) scanning.

Fluorine has a proton number of 9.

F-18 decays to oxygen (O) by β^+ decay.

Write the equation for the decay of a nucleus of F-18 using nuclear notation.

[2]

(b) The β^+ particle (positron) produced travels only a short distance in the patient before it meets an electron and is annihilated.

Calculate the wavelength λ of gamma photons produced.

$\lambda = \dots\dots\dots\text{m}$ [3]

(c) X-rays and gamma-rays are produced by different physical processes.

Briefly describe both processes.

----- [2]

(d) F-18 has a half-life of 109.7 minutes.

Explain the advantage that this has for the patient but the disadvantage that this has for the radiographers.

----- [3]

*Here is some data for a nucleus of carbon-14 ($^{14}_6\text{C}$) and a nucleus of uranium-235 ($^{235}_{92}\text{U}$).

	Carbon-14 nucleus	Uranium-235 nucleus
Decay mode	Beta-minus decay	Alpha decay
Mass of nucleus / u	14.0	235.0
Radius of nucleus / 10^{-15} m	2.9	7.4

Use the data to:

- describe the composition of the nuclei before and after the decay in terms of hadrons and quarks
- show that both nuclei have the same density.

[illegible]

[6]

5 Which of the following statement(s) correctly describe radioactive decay?

- 1 Radioactive decay can be modelled using dice.
- 2 Radioactive decay of nuclei is random.
- 3 Radioactive decay of nuclei is spontaneous

- A Only 1
- B Only 2
- C 2 and 3
- D 1, 2 and 3

Your answer

☐

[1]

6 The half-life of fluorine-18 isotope is T .
After time $t = 4T$ the number of fluorine-18 nuclei in a source is N .

How many fluorine-18 nuclei have decayed in the time interval from $t = 0$ to $t = 4T$?

- A $3N$
- B $4N$
- C $15N$
- D $16N$

Your answer

☐

[1]

- 7 The activity of an alpha-emitting source is 120 kBq. The kinetic energy of each alpha-particle is 4.0 MeV.

What is the rate of energy released by the source?

- A $6.4 \times 10^{-13} \text{ W}$
- B $4.8 \times 10^{-8} \text{ W}$
- C $7.7 \times 10^{-8} \text{ W}$
- D $1.2 \times 10^5 \text{ W}$

Your answer

[1]

- 8 Fig. 24 shows two horizontal metal plates in a vacuum.

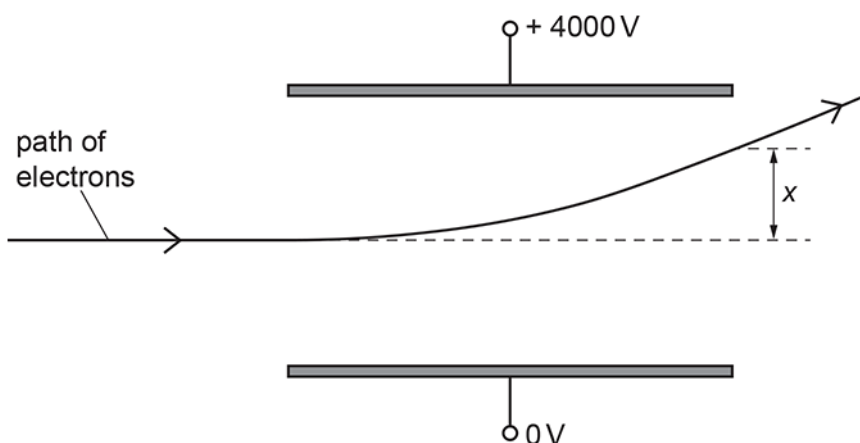


Fig. 24

The arrangement shown in Fig. 24 is now used to investigate positrons emitted from a radioactive source. The speed of the positrons is also $6.0 \times 10^7 \text{ m s}^{-1}$.

The initial path of the positrons is the same as that of the electrons in Fig. 24.

On Fig. 24, sketch the path of the positrons between the plates.

[2]

- 9 A researcher is doing an experiment on a radioactive solution in a thin **glass** tube.
The solution has two radioactive materials **X** and **Y**.
The table below shows some data on these two materials.

	Material X	Material Y
Half-life	10 minutes	10 hours
Particles emitted	Alpha	Beta-minus
Daughter nuclei	Stable	Stable

The solution has the same number of nuclei of **X** and **Y** at the start.

- i. State and explain which material has the greatest activity at the start.

----- [1]

- ii. State why it is dangerous for the researcher to handle the test tube with bare hands.

----- [1]

- 10 A radiation detector is placed in front of a beta-emitting source.
The count-rate is measured and recorded every 10 minutes.
The results are shown below.

311 s⁻¹ 309 s⁻¹ 299 s⁻¹ 307 s⁻¹ 321 s⁻¹

What term can be used to describe the data shown?

- A exponential
B linear
C random
D spontaneous

Your answer

[1]

The nucleus of thorium-232 (${}^{232}_{90}\text{Th}$) emits two alpha particles and two beta-minus particles to become a nucleus of an isotope of radium.

What is the nucleon number A and the proton number Z for the nucleus of this radium isotope?

- A $A = 224, Z = 88$
- B $A = 228, Z = 86$
- C $A = 224, Z = 84$
- D $A = 228, Z = 88$

Your answer

[1]

12 The half-life of the isotope carbon-14 is 5700 years (y).

- i. Show that the decay constant λ for this isotope is about $1.2 \times 10^{-4} \text{ y}^{-1}$.

[1]

- ii. Carbon-dating is a technique used to date an ancient wooden axe.

The ratio of carbon-14 to carbon-12 in the axe material is 78% of the current ratio of carbon-14 to carbon-12 in a living tree.

Calculate the age in years of the wooden axe.

age = y [3]

- iii. State **one** assumption made in the calculation in (ii).

----- [1]

- 13 A student is modelling the decay of a radioactive source using the equation $\Delta N / \Delta t = -0.5 N$.
 The student decides to use $\Delta t = 0.10$ s.
 The number N of radioactive nuclei is 2000 at $t = 0$.

Part of the modelling spreadsheet from the student is shown below.

t / s	Number N of radioactive nuclei remaining at time t	Number of nuclei decaying in the next 0.10 s
0	2000	100
0.10	1900	
0.20		
0.30		

What is the value of N at $t = 0.30$ s?

- A 1700
- B 1710
- C 1715
- D 1805

Your answer

[1]

14 The chemical composition of ancient rocks found on the Earth can be used to estimate the age of the Earth.

Nuclei of rubidium-87 ($^{87}_{37}\text{Rb}$) decay spontaneously into nuclei of strontium-87 ($^{87}_{38}\text{Sr}$).

The half-life of rubidium-87 is 49 billion years.

i. Name the two leptons emitted in the decay of a rubidium-87 nucleus.

1. _____

2. _____

[1]

ii. The percentage of rubidium left in a sample of an ancient rock is 95%.

Estimate the age of the Earth in billion years.

age = billion years [3]

15(a)

An isotope of polonium-213 ($^{213}_{84}\text{Po}$) first decays into an isotope of lead-209 ($^{209}_{82}\text{Pb}$) and this lead isotope then decays into the stable isotope of bismuth (Bi).

Fig. 24 shows two arrows on a neutron number N against proton number Z chart to illustrate these two decays.

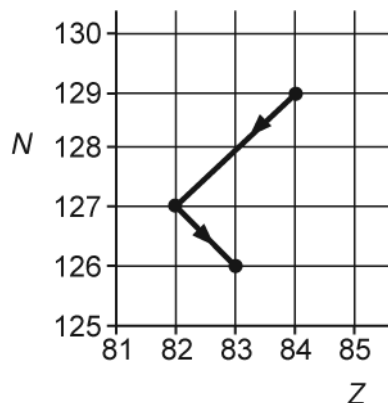
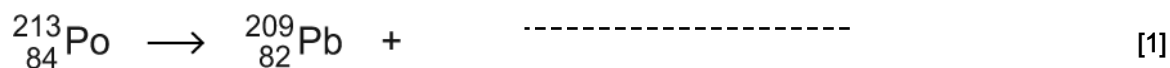


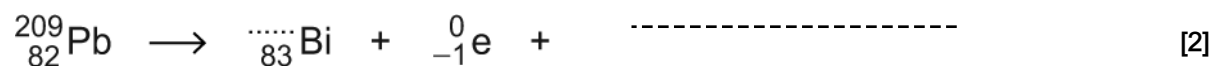
Fig. 24

Complete the nuclear decay equations for

i. the polonium isotope



ii. the lead isotope.



(b) A pure sample of polonium-213 is being produced in a research laboratory.

The half-life of $^{213}_{84}\text{Po}$ is very small compared with the half-life of $^{209}_{82}\text{Pb}$.

After a very short time, the ionising radiation detected from the sample is mainly from the beta-minus decay of the lead-209 nuclei.

i. Briefly describe and explain an experiment that can be carried out to confirm the beta-minus radiation emitted from the lead nuclei.

----- [2]

ii. The activity of the sample of $^{209}_{82}\text{Pb}$ after 7.0 hours is 12 kBq.

The half-life of $^{209}_{82}\text{Pb}$ is 3.3 hours.

Calculate the initial number of lead-209 nuclei in this sample.

number of nuclei = ----- [4]

- (b) The half-life of fluorine-18 is 110 minutes.

Calculate the time t in minutes for the activity of the radiopharmaceutical to decrease to 30% of its initial activity.

$t =$ _____ minutes [3]

- (c) PET scanners are not available in all hospitals. This is because fluorine-18 requires expensive on-site particle accelerators and fluorine-18 has a very small 'shelf-life'.

Suggest the impact this may have on the treatment and diagnosis of patients in the country.

----- [1]

- 18 The medical tracer technetium-99m is used in gamma scans. Technetium-99m has a half-life of 6.0 hours and it emits gamma rays.

A fresh sample of a radiopharmaceutical containing technetium-99m is prepared in the radiography department of a hospital. The initial activity of the radiopharmaceutical is 820 MBq. The radiopharmaceutical is injected into the patient some time later when its activity has dropped to 630 MBq.

Calculate the time in hours between the radiopharmaceutical being produced and it being injected into the patient.

time = _____ h [3]

- 19 The table shows data on four freshly prepared radioactive samples A, B, C and D.

Sample	Number of active nuclei in the sample	Half-life of the sample
A	N	T
B	N	$3T$
C	$5N$	$0.5T$
D	$8N$	$4T$

Which sample has the **smallest** activity?

Your answer

[1]

- 20(a) State what is meant by the *decay constant* of an isotope.

----- [1]

- (b) A radioactive substance has 2000 nuclei. The decay constant of the isotope of the substance is 0.10 s^{-1} .

Use the equation $\frac{\Delta N}{\Delta t} = -\lambda N$ and $\Delta t = 1.0 \text{ s}$ to estimate the number of nuclei left after time $t = 2.0 \text{ s}$.

number of nuclei left = ----- [2]

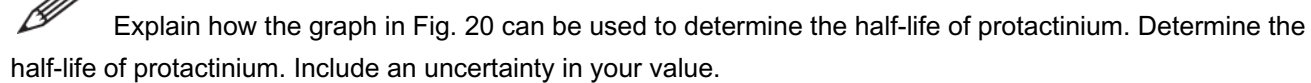
- (c) * A group of students are investigating the decay of protactinium.

A fresh sample of protactinium is prepared. The activity of the sample was measured at intervals of 1.0 minutes for 6.0 minutes.

The table shows the activity corrected for background radiation.

time t / min	0	1.0	2.0	3.0	4.0	5.0	6.0
activity A / Bq	943	523	287	161	79	61	20

Fig. 20 shows the variation of $\ln(A)$ with time t .



[6]

(b) Fluorodeoxyglucose (FDG) is a radioactive tracer often used for PET scans. It contains radioactive fluorine-18, which is a positron-emitter. Some information about FDG and fluorine-18 is given below.

- 9.9% of the mass of FDG is fluorine-18.
- The half-life of fluorine-18 is 6600 s.
- The molar mass of fluorine-18 is $0.018 \text{ kg mol}^{-1}$.

A patient is injected with FDG. The initial activity of FDG is 400 MBq.

Use the information given to calculate the initial mass of FDG given to the patient.

mass = _____ kg [4]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a	i	$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{6.01 \times 3600} (= 3.20 \times 10^{-5} \text{ s}^{-1})$ $(A = \lambda N) \quad 900 \times 10^6 = 3.20 \times 10^{-5} \times N$ $N = 2.8 \times 10^{13}$	C1 C1 A1	Correct to at least 2sf (2.81) Special case: Calculation of decay constant in hour^{-1} gives 0.115 leading to $N = 7.8 \times 10^9$ maximum 1 mark Calculation of decay constant in min^{-1} gives 1.9×10^{-3} leading to $N = 4.7 \times 10^{11}$ maximum 1 mark <u>Examiner's Comments</u> It was very encouraging to see that nearly two thirds of candidates were able to score all 3 marks on this question. Logarithms can challenge candidates but it was clear that many understood how to carry this out and were able to effectively. There were relatively few who were confused by the time units, but credit could be given even when the unit was used incorrectly.
		ii	(Correct use of $A = A_0 e^{\lambda t}$) $0.03 = e^{-3.2 \times 10^{-5} t}$ $\ln(0.03) = -3.2 \times 10^{-5} t$ $t = 109\,500 \text{ s} = 30 \text{ hours}$	C1 C1 A1	Or use of λ as 0.115 hour^{-1} ecf for λ from (b)(i) $\ln(0.03) = -0.115t$ Allow final answer that rounds to 30 or 31 hours Special case: if fall of 3% ($A = 0.97A_0$) calculated, leading to 0.26 hours then max 1 <u>Examiner's Comments</u> This calculation was also carried out very well, with many gaining full marks. There are several different ways to obtain the correct answer and part credit can be given for any method which will use the decay equation correctly. Most candidates calculated what 35 of the initial activity would be, putting both A and A_0 into the calculation, rather than just using 0.03 as the ratio.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	Activity is low compared to background	B1	Allow activity is very low Allow decays very slowly / very little decay for low activity Allow rate of emission for activity Ignore references to long half life <u>Examiner's Comments</u> The required response here was to do with the activity of the remaining Tc. It needed to be clear that the activity was very low, or that it was low compared to any background count. Using the half-life alone as the response would not be worth any credit. A noticeable number of candidates misinterpreted the question and answered why the presence of the Tc does continue to cause a risk. Many also gave responses relating to the beta emission, stating that it was safe or wouldn't affect bones.
	c	Level 3 (5–6 marks) Clear description and full correct calculation <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description or full correct calculation OR Some description and some calculation <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or limited calculation <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark No response or no response worthy of credit	B1×6	Use level of response annotations in RM Assessor Indicative scientific points may include: Expect to see appropriate safety precaution for clear description Description: • Take background count • Over (at least) 3 minutes • Take count due to source • Over the same time • Measured using a stopwatch • Calculate the corrected count • Calculate the corrected count per minute • Use a Geiger-Muller tube and counter • Hold source at arms length when transferring • Keep in lead lined box at other times • Use tongs to hold source Calculation • $C = C_0 e^{-\lambda t}$ • $\ln C = \ln C_0 - \lambda t$ • e.g $7.51 = 7.81 - \lambda \times 6$ • λ around 0.05 hour^{-1} • $T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$ • $T_{\frac{1}{2}} = 13.9 \text{ hours}$ • (allow any answer that round to 14 hours)

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			• $T_{\frac{1}{2}} = 830$ minutes or 49 900 seconds (allow some tolerance for slight graph reading discrepancies) • Or by a graphical method Examiner's Comments This is the second Level of Response (LoR) question in this paper, and was designed to assess nuclear and particle physics, specifically sections 6.4.3e of the specification. In particular, there are two parts (description and calculation) relating to a method and graphical data. As previous, there are no specific marking points and examiners use a 'best-fit' approach to the marking. Indicative scientific points are given in the mark scheme as a guidance what to expect, but not all need to be satisfied to be given full marks. Incorrect physics can be expected to be penalised, however. Candidates should always be encouraged to structure their answers carefully and in calculations aim to give explanation of each stage rather than just producing a number. The two separate sections can be separated into a description of the method for obtaining the data, and a calculation using the graph to determine the half-life. The key points in the description that examiners were looking for were: - named apparatus - measurement of background radiation - calculation of corrected count rate - safety requirements. The key points in the calculation that examiners were looking for were: - arrangement of decay equation having taken logs - demonstration that the gradient is the decay constant - calculation of the gradient - use of the gradient to determine the half-life. The descriptions of the method varied significantly. Some were extensive and detailed and others a very limited response despite being a procedure that candidates are likely to have come across. Most candidates gave a simple safety precaution, such as holding at arm's length, but there were plenty of vague responses such as wearing coats. Not all

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					candidates could explain how, or why, the background was taken and a fair number missed this out completely. Quite a few candidates gave extensive detail about storing the source, which although is relevant, is not the key point to the question. The analysis was done better and in general well-structured and clear. The working was set out well for the most part and it was simple to follow the candidates working. Not all gave a clear reason for how they were obtaining the half-life, but this was not fully necessary as long as the answer was correct. One common error was to take logs of the data on the y-axis rather than appreciating that they are already log values. There were not many misreads, but the nature of the data meant that any could have a significant impact on their answer and would likely be penalised. Some candidates got into difficulties with time units and attempted to convert the decay constant to s^{-1} without success.
			Total	13	
2			B	1	<u>Examiner's Comments</u> A little under half of the candidates were able to correctly identify the correct statements. A helpful approach to this type of question is to 'tick' the statements which the candidate is certain of, which then reduces the options available.
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
3	a	${}^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^0_1\beta^+ (+\nu)$	C1 A1	Correct fluorine isotope Correct equation ALLOW numbers written to right of symbol ALLOW $e / e^+ / \bar{e} / \beta$ for positron symbol with correct numbers IGNORE gamma in products ALLOW ${}^0_0\nu$ for neutrino ALLOW ν_e for neutrino <u>Examiner's Comments</u> Most candidates will have gained the first mark for the correct isotope of fluorine. Those who didn't likely reversed the positions of the nucleon and proton number. It was noticeable that a large number of candidates were unable to balance the equation and while they had the correct values for the positron, the oxygen isotope was incorrect.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$E = mc^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2$ $= 8.2 \times 10^{-14} \text{ (J)}$ $E = hc / \lambda = 8.2 \times 10^{-14} = 6.63 \times 10^{-34} \times 3 \times 10^8 / \lambda$ $\lambda = 3.00 \times 10^8 \div 1.2367 \times 10^{20} = 2.4 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Working leading to evaluation of E ALLOW factor of 2 leading to 1.6×10^{-13} (J) NOT use of $\lambda = h/mv$ where v is 3×10^8 alone XP ALLOW use of $\lambda = hc/mc^2$ or $\lambda = h/mc$ ALLOW Use of electron mass = 0.511 MeV giving 8.2×10^{-14} (J) Correct to at least 2 significant figures Answer of 4.9×10^{-12} or 1.2×10^{-12} allow 2/3 due to incorrect accounting for factor of 2 <u>Examiner's Comments</u> Well over half of the candidates were able to score 2 or more marks on this question. A great deal scored 2 marks due to the inclusion of a factor of two, which was then not removed (or vice versa). Some candidates used the formula for kinetic energy – which by coincidence may give the correct answer – however this was not credited, nor was the more common use of the de Broglie formula. Examiners sometimes had a difficult decision on this question, whether the response was a physics error or not, and if in doubt a candidate would be awarded the marks.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	c	X-rays formed when <u>electrons</u> (in an atom) de-excite Gamma rays come from the decay of <u>nuclei</u> (in unstable isotopes)	B1 B1	ALLOW X-rays may be produced by acceleration / deceleration of (fast moving) electrons / X-rays are produced when (fast moving) electrons are incident on a metal target / X-rays may be produced when electrons lose kinetic energy ALLOW gamma rays come from (the decay of) radioactive <u>nuclei</u> / gamma rays come from the <u>nucleus</u> of unstable atoms / gamma rays come from the de-excitation of <u>nuclei</u> / gamma rays come from <u>annihilation</u> of particle-antiparticle (pairs) / IGNORE gamma from fission <u>Examiner's Comments</u> The main principle behind this question was to distinguish between the X-ray production by electrons and gamma ray production by nuclei. Fairly specific details were required for each, although a wide range of answers were accepted. This question discriminated well among the candidates with roughly equal fractions getting 0, 1 and 2 marks.
	d	The half-life is short Advantage: Exposure of the patient to <u>ionising</u> radiation is kept as low as possible. Disadvantage: (Radiographers have a) short time to scan/diagnose the patient	B1 B1 B1	ALLOW activity is high IGNORE it decays quickly ALLOW F18 has to be manufactured on site before use / high activity means exposure is high during handling IGNORE short time to treat the patient <u>Examiner's Comments</u> Although apparently a simple set of ideas, the detail required for this question meant that only a small fraction of candidates were able to gain full marks. It was necessary to state that the half-life was (relatively) short, which many of the stronger candidates did not do. The advantage to the patient requires the use of the term 'ionising' as this is the fundamental issue with the radiation. Similarly, candidates had to make it clear that the disadvantage was due to the short time to carry out the scan rather than just a statement of needing to work quickly.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	10	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
4	Level 3 (5–6 marks) Clear description including at least one correct comment on quarks and analysis of density <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Some description and some analysis of density or clear description or clear analysis of density <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or Limited analysis <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks <i>No response or no response worthy of credit.</i>	B1 × 6	Indicative scientific points may include: Description (max L2 if either alpha or beta not described) ${}^{14}_6\text{C} \rightarrow {}^0_{-1}\text{e} + {}^{14}_7\text{X} \text{ (ignore identification of X)}$ • Before: 6 protons and 8 neutrons • After: 7 protons and 7 neutrons • Neutron → u d d and/or Proton → u u d • Neutron changes into proton (+ electron + antineutrino) • Down quark changes into an up quark (+ electron + antineutrino) ${}^{235}_{92}\text{U} \rightarrow {}^4_2\alpha + {}^{231}_{90}\text{Y} \text{ (ignore identification of Y)}$ • Before: 92 protons and 143 neutrons • After: 90 protons and 141 neutrons • 2 neutrons and 2 protons removed (by alpha particle) • Has 6 fewer u quarks and 6 fewer d quarks Density (max L2 if analysis done without use of data) • masses: 14.0(u) and 235.0(u) • ratios: masses = 16.8; radii = 2.55; volumes = 16.6 • volume ~ 2.9³ and 7.4³ (ignore 4/3π and 10⁻¹⁵) • densities calculated (about 2.3 × 10¹⁷ kgm⁻³) • density shown to be (roughly the) same Examiner's Comments This is the second level of response (LoR) question in this paper, and was designed to assess nuclear and particle physics, specifically sections 6.4.1 and 6.4.2 of the specification. In particular, there are two parts (description and analysis) relating to two radioisotopes, and some given data. As previous, there are no specific marking points and examiners use a “best-fit” approach to the marking. Indicative scientific points are given in the mark scheme as a guidance what to expect, but not all need to be satisfied to be given full marks. Incorrect physics can be expected to be penalised however. Candidates should always be encouraged to structure their answers carefully and in calculations

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			aim to give explanation of each stage rather than just producing a number. The two separate sections can be separated into a description of the decays, and an analysis of the data to show the two nuclear densities are the same. The key points in the description that examiners were looking for were:—numbers of protons before and after each decay – a description of the change in quark structure during beta decay – correctly balanced equations for alpha and beta-minus decay – the nature of the emissions in each decay. The key points in the analysis that examiners were looking for were: – calculation of volumes of nuclei, using $(4/3\pi) r^3$ – correct use of the density formula – numerical evaluation of the densities of the nuclei and a comparison. The descriptions of the decays varied hugely both in quality and quantity. Most candidates were aware that a beta particle is an electron, but several seemed to think it was an orbiting one which had no structural changes on the nucleus. Those who did know it was from a neutron to proton change often mixed up the nucleon numbers afterwards. Alpha decay was generally done better with most candidates knowing the nature of an alpha particle and correctly stating the structure of the remaining nucleus. Attempts were generally made to changes in quark structures (mostly for beta decay) and the structure of a neutron and proton were known, although strangely several candidates stated the change was from an up quark to a down quark. Candidates who did not give any descriptions of the quark structures were limited to Level 2. For the most part, the analysis was done better than the description and candidates were generally able to correctly evaluate the two densities and make a suitable comparison. Units of density were not penalised here, with many candidates using the mass (in u) in the calculation but giving the units of density as kg m^{-3}, although many stating u m^{-3}. However, many did correctly calculate the mass of the nucleus using the correct factor

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			coming out with densities of approximately $2.3 \times 10^{17} \text{ kg m}^{-3}$. Those who took an approach to algebraically show that the density of every nucleus is independent of the nucleon number (using a standard method to get to some variation of $3u/4\pi r^3$) alone were unable to reach Level 3 as they had not used the data. The use of r_0 as $1.2 \times 10^{-15} \text{ m}$ was commonly seen. ${}^6_6\text{C} \rightarrow {}^6_7\text{N} + {}^0_{-1}\text{e} + \bar{\nu}_e$ beta $\rightarrow$ neutron neutron $\rightarrow$ proton neutron = UDD & proton = UUD UDD In the beta minus decay, 1 down quark turns to a up quark and one proton turns into a proton A candidate who provides some description and analysis which places it in Level 2, but the confusion over alpha decay and general lack of clarity means it would be at the lower point (3 marks).
	Total	6	

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5			D	1	<u>Examiner's Comments</u> This question proved to be relatively straightforward with most candidates selecting the correct response. By far the most common incorrect response was C, suggesting that the idea of modelling using dice was not known, despite it being specification point 6.4.3f(ii).
			Total	1	
6			C	1	<u>Examiner's Comments</u> Only around one third of candidates were able to give the correct response of C. Many candidates showed a variety of correct working to produce the correct initial number of nuclei as 16 N, however then gave the incorrect response of D. This highlights the need to read each question carefully.
			Total	1	
7			C	1	<u>Examiner's Comments</u> Although a little over two thirds of candidates selected the correct response, this question discriminated well between candidates of differing abilities. Potential errors in this question included two unit prefix conversions and the conversion between eV and joules. Incorrect responses were fairly well spread across the other distractors.
			Total	1	

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8			Downward curved path	B1	Ignore any line outside of the plates
			Same x	B1	Expect same x by eye <u>Examiner's Comments</u> Nearly all candidates appreciated that the path should be downwards, but many did not take the care needed for it to be clear that the deflection at the end of the plate should be the same. Some candidates drew an 'x' on their sketch, which was helpful in determining if the intention to draw it the same had been made.
			Total	2	
9		i	Material X because of the shorter half-life	B1	Must be comparative Allow explanation in terms of decay constant <u>Examiner's Comments</u> This question expects the candidates to appreciate that the activity is related to the half-life. The majority of candidates were able to successfully answer this question although a number did not make it comparative and simply said that X had a short half-life.
		ii	(Alpha particles are stopped by the glass but) the beta-particles are not (AW)	B1	Allow symbols <u>Examiner's Comments</u> Not many candidates recognised that the penetrating powers of the radiations through glass were required for the response; most referred to the ionising (and so harmful to health) properties of both sources.
			Total	2	

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10			C	1	<u>Examiner's Comments</u> The correct response is C. The responses are terms used frequently when studying data, but as around only one half of the candidates were able to get the correct response, it is clear that they are not fully understood. A little perplexingly, the most common incorrect response was B, as it is difficult to see how this data could be considered linear. This does show how easy it is to assume that candidates are confident in their use of terminology, just because they are frequently used.
			Total	1	
11			A	1	<u>Examiner's Comments</u> The correct response is A. Around half of candidates were able to select the correct response. Although it would seem appropriate to write out some simple decay sequence, many candidates showed little working here. Some were able to get the correct response (probably through mental arithmetic) but incorrect responses may simply have been down to a lack of knowledge of nuclear changes, most likely in the beta decay. Incorrect responses were spread fairly evenly among the distractors, again suggesting that this topic was not well understood.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
12		i	$(\text{decay constant} =) \frac{\ln 2}{5700}$ $\text{decay constant} = 1.2(2) \times 10^{-4} \text{ (y}^{-1}\text{)}$	C1 A0	
		ii	$0.78 = e^{-\lambda t}$ $\ln 0.78 = (-) 1.2 \times 10^{-4} \times t$ $\text{age} = 2100 \text{ (y)}$	C1 C1 A1	Note $1 = 0.78e^{-\lambda t}$ is XP; answer is negative (– 2100 y) There is no ECF from (b)(i) Note 1.22×10^{-4} gives an answer of 2040 y or 2000 y
		iii	The ratio (of carbon-14 to carbon-12) has remained constant	B1	
			Total	5	
13			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
14		i	Electron and (electron) antineutrino	B1	Allow beta-minus (particle) / β^- ; $\bar{\nu}_{(e)}$ Allow anti electron neutrino <u>Examiner's Comments</u> The majority of the candidates scored a mark for electron and antineutrino as the two leptons. The most common incorrect answers were <i>neutrino</i>, <i>positron</i>, <i>proton</i> and <i>neutron</i>. The pairing of electron and positron also appeared on some scripts.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	$\lambda = \ln 2 / 49$ or $\lambda = 0.0141$ (billion y^{-1}) $0.95 = e^{-0.0141t}$ or $0.95 = e^{-4.48 \times 10^{-19}t}$ (age = $-\ln(0.95)/0.0141$) age = 3.6 (billion years)	C1 C1 A1	Allow $\frac{\ln 2}{49 \times 10^9 \times 3.16 \times 10^7}$ or $4.48 \times 10^{-19} \text{ (s}^{-1}\text{)}$ Allow both C1 marks for $\ln(0.95) = \ln(0.5) \times t/49$ Allow $0.05/0.0141 \approx t$ (this gives 3.5.. for the final mark) Note age in seconds is $1.15 \times 10^{17} \text{ (s)}$; this will score 2 marks Examiner's Comments Many of the top-half candidates demonstrated how the age of the Earth could be calculated in just a few lines. The use of natural logs ($\ln$) was faultless. Most candidates calculated the decay constant and then used the equation $0.95 = e^{-\lambda t}$, or its equivalent $\ln 0.95 = -\lambda t$, to calculate the age t. Candidates are reminded not to round numbers in long calculations – it is good practice to keep all the digits on your calculator. A significant number of candidates rounded the decay constant to 2 SF ($0.014 \text{ billion } y^{-1}$), and this gave an answer of 3.7 billion years. The correct answer, without rounding $\ln 2/49$, was 3.6 billion years. On this occasion, examiners allowed the 3.7 billion years answer. The command term 'estimate' in the question made a small number of candidates to use the equation $\frac{\Delta N}{\Delta t} \approx -\lambda t$. This was allowed, and it gave an estimated age of 3.5 billion years.  Misconception These were some common errors being made in this question, these are summarised below. • Incorrect conversion of billion years into seconds. (Most candidates calculated the decay constant in billion y^{-1}, which easily led to the correct answer in billion years.) • Using $\ln 0.05$ instead of $\ln 0.95$ when calculating the age of the Earth.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
15	a	i	alpha-particle / ${}^4_2\text{He}$ / ${}^4_2\alpha$	B1	
		ii	nucleon number for Bi = 209 antineutrino / ${}^{(0)}_{(0)}\bar{\nu}_{(e)}$	B1 B1	Note: Do not allow incorrect subscript and superscript
	b	i	Aluminium (sheet placed between source and detector) The count (rate) reduces or Magnetic / electric field used Electrons identified from correct deflection / motion in field	M1 A1 M1 A1	Allow count (rate) drop to background / zero Allow 2 marks for 'the range in air is a few m' Examiner's Comments This turned out to be a low-scoring question from candidates across the ability spectrum. Only a quarter of the candidates gained 2 marks for identifying aluminium as the absorber for the beta-minus radiation (electrons) and providing adequate description in terms of reduction in the count-rate. A small number of candidates opted for charged parallel plates and identified the electrons curving towards the positive plate. There were some baffling descriptions involving pointing the source at '<i>wires and measuring the current</i>'. Fluorescent screens and cloud chambers were not allowed as acceptable answers because both can be used to detect the presence of gamma-photons and alpha-particles.

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Question			Answer/Indicative content	Marks	Guidance
		ii	$(\lambda =) \ln 2 / 3.3 \text{ (h}^{-1}\text{) or } (\lambda =) 0.21 \text{ (h}^{-1}\text{)}$ $(A_0 =) 12 \times 10^3 / e^{-(0.21 \times 7.0)} \text{ or } (A_0 =) 5.219 \times 10^4 \text{ (Bq)}$ $(N_0 =) 5.219 \times 10^4 / 5.835 \times 10^{-5}$ number of nuclei = 8.9×10^8 Or $(\lambda =) \ln 2 / [3.3 \times 3600] \text{ (s}^{-1}\text{) or } (\lambda =) 5.835 \times 10^{-5} \text{ (s}^{-1}\text{)}$ $(N =) 1.2 \times 10^4 / 5.835 \times 10^{-5} \text{ or } 2.057 \times 10^8$ $(N_0 =) 2.057 \times 10^8 / e^{-(0.21 \times 7.0)}$ number of nuclei = 8.9×10^8	C1 C1 C1 A1 C1 C1 C1 A1	Allow credit for alternative methods Note this is the same as $12 \times 10^3 \div (0.5)^{7.0/3.3}$ Note 9.0×10^8 can score full marks if numbers are rounded Possible ECF for incorrect conversion of time Note this is the same as $2.057 \times 10^8 \div (0.5)^{7.0/3.3}$ <u>Examiner's Comments</u> The question was multi-stepped calculation, requiring knowledge of radioactive decay equations, half-time and activity. The final stage of the calculation was dependent on the equation $A = \lambda N$ and working consistently in Bq for the activity and in s^{-1} for the decay constant. The number of nuclei N could not be calculated with the activity in Bq and the decay constant in either h^{-1} or min^{-1} . About half of the candidates scored full marks. Those working with inconsistent units invariably ended up with the incorrect value 2.5×10^5 nuclei, but this still earned them 2 marks for the preceding steps.
			Total	9	

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Question			Answer/Indicative content	Marks	Guidance
16			A	1	<u>Examiner's Comments</u> The majority of the candidates did get the correct answer A. A significant number of candidates opted for C, confusing contrast material with medical tracers.
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
17	a	The patient is surrounded by (gamma) detectors or Increased activity is where F-18 accumulates (AW)	B1	Allow 'diametrically opposite detectors'
		The positrons (from the F-18) <u>annihilate</u> electrons (inside the patient)	B1	
		Each annihilation produces two gamma photons travelling in <u>opposite</u> directions	B1	Not gamma rays / radiation
		The arrival times are used to locate position (of increased activity)	B1	Allow 'delay time' Examiner's Comment Most candidates scored two or more marks for their description of the PET scanner. Most candidates knew that the annihilation of positrons and electrons was central to the scanning technique. A small number of candidates either confused the PET scanning with CAT scanning or assumed that the gamma detectors were monitoring the emission of positrons from the patient.
	b	$\lambda = \ln 2 / 110$ or $6.3 \times 10^{-3} \text{ (min}^{-1}\text{)}$	C1	Allow $1.05 \times 10^{-4} \text{ (s}^{-1}\text{)}$
		$0.30 = e^{-6.3 \times 10^{-3} t}$		This is the same as $0.30 = e^{-1.05 \times 10^{-4} t}$
		$t = \frac{\ln(0.30)}{-6.3 \times 10^{-3}}$	C1	Note: This mark is for a ln expression (any subject)
		$t = 190 \text{ (minutes)}$	A1	Allow 2 marks for $1.15 \times 10^4 \text{ (s)}$ as the final answer Examiner's Comment This was not an easy question. It required knowledge and understanding of activity, decay constant and natural logs. It is good to report that most of the candidates produced immaculate answers. The common mistakes made were: - Using either $\ln(1/3)$ or $\ln(0.70)$ rather than $\ln(0.30)$ in the calculations. - Assuming the decay was linear rather than exponential.

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	c		Any sensible suggestion, e.g. 'post-code' lottery, some patients may not get the treatment because of where they live, longer waiting lists, etc.	B1	Examiner's Comment Almost all candidates gave a plausible suggestion in this last question in the paper. It is good to report that physicists are mindful of the impact of science on society.
			Total	8	
18			$\lambda = \ln 2 / 6.0 = 0.116 \text{ (h}^{-1}\text{)}$ $(A = A_0 e^{-\lambda t})$ $t = \frac{\ln(630/820)}{0.116}$ $t = 2.3 \text{ (h)}$	C1 C1 A1	Allow $\lambda = \ln 2 / (6.0 \times 3600) = 3.21 \times 10^{-5} \text{ (s}^{-1}\text{)}$ Allow $\frac{\ln(630/820)}{3.2 \times 10^{-5}} \text{ (= 8200 s)}$
			Total	3	
19			B	1	
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
20	a		The decay constant is the probability of decay of a nucleus per unit time.	B1	Allow: the decay constant is the fraction of nuclei decaying per unit time. Allow: decay constant = activity ÷ number of nuclei left in a sample.
	b		number decaying in 1st second = $2000 \times 0.10 = 200$ number decaying in the 2nd second = $1800 \times 0.10 = 180$ number left = $1800 - 180 = 1620$	C1 A1	
	c		Level 3 (5–6 marks) Correct explanation Correct determination of λ and half-life Correct determination of uncertainty (Maximum 6 marks) Any point omitted or incorrect (5 marks). <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Mostly correct explanation Mostly correct determination of λ and half-life Some attempt of determining uncertainty (Maximum 4 marks) Any point omitted or incorrect (3 marks). <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Basic explanation Some attempt to determine λ or half-life No attempt at uncertainty. (Maximum 2 marks) <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1 x 6	Explanation $A = A_0 e^{-\lambda t}$ $\ln A = \ln A_0 - \lambda t$ - A graph of $\ln A$ against t will be a straight line with gradient $(-)\lambda$ - half-life = $\ln 2 / \lambda$ Determination - Line of best fit drawn - Gradient determined using a large triangle - decay constant in the range 0.5 to 0.7 min^{-1} - half-life in the range 1.0 to 1.4 min Uncertainty - Worst line of fit drawn - Correct attempt to determine uncertainty
			Total	9	

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Question		Answer/Indicative content	Marks	Guidance
21	a	* Level 3 (5–6 marks) For equipment expect both E1 and E2 Descriptions has all the points At least two safety precautions mentioned Both Q1 and Q2 mentioned for the quality of results. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Expect at least E1 for equipment For description expect D1 and D2 and an attempt at either D3 or D4 At least one safety point mentioned Expect either Q1 or Q2 for quality of results. <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Expect at least E1 for equipment For description expect D1 and D2 At least one safety point mentioned Statements for quality are not relevant. <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1	Equipment (E) 1. GM tube, counter or rate-meter and lead plates used 2. Micrometer or vernier calliper (to measure thickness of plates). Description (D) 1. Measure counts for a specific time and hence the count-rate for each thickness of lead 2. Vary the thickness of lead and record the count-rates 3. Plot a graph of count-rate against thickness and determine the half thickness of lead 4. Fig. 23.1 is used to determine the photon energy. Safety (S) 1. Do not point source at person 2. Keep safe distance between you and source 3. Use tongs to handle source. Quality of results (Q) 1. The counts are recorded over a long period of time 2. Background radiation taken into account.
	b	$\lambda = \frac{\ln 2}{6600} = 1.050 \times 10^{-4} \text{ (s}^{-1}\text{)}$ $N = \frac{400 \times 10^6}{1.050 \times 10^{-4}} = 3.809 \times 10^{12}$ mass of FDG = $\frac{3.809 \times 10^{12}}{6.02 \times 10^{23}} \times 0.018 \div 0.099$ mass of FDG = $1.15 \times 10^{-12} \text{ (kg)}$ or $1.2 \times 10^{-12} \text{ (kg)}$	C1 C1 C1 A1	Correct use of $A = \lambda N$
		Total	10	